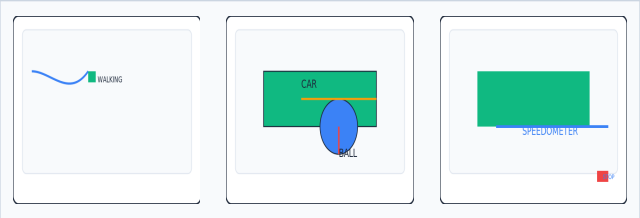
DraftBench — Special Education Resource

Curriculum Aligned & Autism Support Framework



Kinematics Basics

Subject: Physics | **Target Grade:** 11

Autism-Friendly Learning Framework

*Predictable pacing: One core concept per section.

*Literal, concrete language with real-world grounding.

Step 1: Introduction to Kinematics

Core Concept: Study of motion without considering forces

Explanation: Motion, speed, and distance are basic concepts in physics

Key Takeaways:

• Motion describes how objects move

• Speed measures how fast objects move

• Distance is how far objects travel

Real-World Example: Walking to school

Communication Support: point to objects

Vocabulary Builder:

• **Motion:** How objects move

• **Speed:** How fast objects move

• **Distance:** How far objects travel

Step 2: Types of Motion

Core Concept: Different ways objects can move

Explanation: Objects can move in different ways

Key Takeaways:

• Linear motion is straight line

• Circular motion is around a circle

• Rotational motion is spinning

 Real-World Example: Car driving on road

 Communication Support: point to diagrams

 Vocabulary Builder:

• **Linear:** Straight line motion

• **Circular:** Around a circle

• **Rotational:** Spinning

Step 3: Speed and Velocity

 Core Concept: Speed and velocity describe motion

 Explanation: Speed and velocity help describe motion

 Key Takeaways:

• Speed measures how fast

• Velocity includes direction

• Speed is scalar, velocity is vector

 Real-World Example: Car driving at 60 km/h

 Communication Support: point to diagrams

 Vocabulary Builder:

• **Speed:** How fast an object moves

• **Velocity:** Speed with direction

Step 4: Practical Examples

 Core Concept: Real-life examples of speed and velocity

 Explanation: Examples help us understand speed and velocity

 Key Takeaways:

• Car speed on highway

• Ball rolling on a table

• Person walking to school

 Real-World Example: A car driving on a highway and a ball rolling on a table

 Communication Support: point to diagrams

 Vocabulary Builder:

• **Speed:** How fast an object moves

• **Velocity:** Speed with direction

Step 5: Review of Kinematics

 Core Concept: Review of motion, speed, and velocity

 Explanation: Review helps us remember the concepts

 Key Takeaways:

• Motion describes how objects move

• Speed measures how fast

• Velocity includes direction

 Real-World Example: Walking to school

 Communication Support: point to diagrams

 Vocabulary Builder:

• **Motion:** How objects move

• **Speed:** How fast an object moves

• **Velocity:** Speed with direction